Qingkai Dong

Email: qingkai.dong@uconn.edu Website: dqksnow.github.io Phone: +1 (860) 931-9075

Education

University of Connecticut Aug 2023 – Expected 2027
Ph.D. in Statistics Storrs, CT, USA
Advisors: Prof. HaiYing Wang and Prof. Jun Yan Passed the Ph.D. Qualifying Exam

Zhongnan University of Economics and Law Sep 2020 – May 2023
M.S. in Mathematical Statistics Wuhan, China
Advisor: Prof. Hui Zhao
Thesis: Model Averaging and Variable Selection for Accelerated Failure Time Models

Qingdao University Sep 2016 – May 2020
B.S. in Applied Statistics Qingdao, China

Research Experience

Research Assistant (Statistical Methodology) Aug 2023 – Present
University of Connecticut

- **Primary output:** First-authored methodology paper on rare-feature-adaptive subsampling for regression models; manuscript under review.
- Developed theory showing that estimation efficiency for rare binary covariate coefficients is driven by the limited number of rare observations, explaining numerical instability and slow convergence.
- Introduced a rarity-adaptive subsampling method that quantifies feature rarity to ensure adequate representation of rare features, improving robustness and efficiency.
- Implemented the methods in the subsampling R package; validated performance via theoretical results, simulation studies, and real-data applications; produced documentation and reproducible examples.

Research Assistant (Academia–Industry Collaboration) Mar 2024 – Dec 2025
Servier Pharmaceuticals and the University of Connecticut

- **Primary output:** Coauthored a manuscript introducing a predictive-modeling–assisted interim analysis framework for censored time-to-event endpoints; manuscript under revision.
- Developed a covariate-informed approach that augments interim data by predicting event times for censored participants, improving estimation of treatment effects and conditional power.
- Proposed evaluation metrics for conditional power forecast accuracy and futility decision quality; conducted extensive simulations to characterize when performance improves or degrades under varied censoring, covariate informativeness/type, prediction range, and sample size.

Global Biometrics Intern Jun 2026 – Aug 2026
Servier Pharmaceuticals

- Developed a modular workflow to reconstruct synthetic individual participant data from published Kaplan–Meier curves and summary statistics using target-matching loss functions.
- Demonstrated close reproduction of reported survival metrics and evaluated recovery of unreported quantities through simulation studies; applied the workflow to internal data.

Research Assistant (Social Determinants of Health, Frailty, and Accelerated Aging in Breast Cancer Survivors) Jul 2024 – Aug 2024
Departments of Statistics and Human Development and Family Sciences, University of Connecticut

- Helped frame and prepare a study examining social determinants of health, frailty, and accelerated aging among breast cancer survivors.
- Conducted data retrieval, cleaning, and integration across All of Us EHR, survey, and physical-measurement data to support subsequent statistical modeling.

Research Project (Chemical Sensor Data Analytics) Mar 2024 – May 2024
Departments of Statistics and Chemistry, University of Connecticut

- Analyzed fluorescence response data from porphyrinoid sensors; applied clustering and statistical analysis to improve compound differentiation and sensor selection.
- Produced interpretable summaries and visualizations to support experimental decision-making.

Cross-Disciplinary Collaboration 2025 – Present
University of Connecticut & collaborators

- Coauthored a survey on knowledge localization in large language models, distinguishing representational, causal, and editable localization and synthesizing evidence and measurement practices.
- Coauthored a survey on time-series Explainable AI (XAI), organizing methods into signal, pattern, structural, and semantic explanations and reviewing LLM-enabled semantic explanations and benchmark resources.

Software & Open-Source

R package subsampling (CRAN): A statistical computing toolkit for optimal subsampling in big-data modeling. Supports GLMs, quantile regression, and rare event/rare feature regimes.

- Implemented optimized sampling probability computation targeting variance reduction and prediction accuracy, including practical routines for pilot sampling and two-step designs.
- Engineering focus: modular API, documentation/vignettes, reproducible examples, and scalable computation patterns.

Teaching Experience

Principal Instructor, University of Connecticut Spring 2027
STAT 3255: Introduction to Data Science

Teaching Assistant, University of Connecticut Sep 2026 – Dec 2026
STAT 1100Q: Elementary Concepts of Statistics; STAT 3445: Introduction to Mathematical Statistics II

Principal Instructor, University of Connecticut Jan 2026 – May 2026
STAT 3675Q: Statistical Computing. Independently taught three 50-minute sections for approximately 20 students; designed slides and R-based assignments, coordinated student presentations, and graded midterm and final exams. Covered data management, visualization, reproducible workflows, regression, nonparametric inference, GLMs, categorical data, time series, and classification.

Selected Papers

Dong Q, Yan J, Zhang Q, Wang H. Rare-feature-adaptive subsampling for big data regression. *Under review.*

Wang D[†], **Dong Q**[†], Zhang Q, Hu Q, Wang H. Random Subsampling: Inferential Targets, Estimators, and Sampling Procedures. *Under review.* [†]Equal contribution.

Dong Q[†], Man A[†], Hua Z, Teng Z, Wang H, Yan J. Predictive-Modeling–Assisted Interim Decision-Making in Adaptive Trials with Censored Survival Outcomes. *Under revision.* [†]Equal contribution.

Dong Q, Liu B, Zhao H. Weighted Least Squares Model Averaging for Accelerated Failure Time Models. *Computational Statistics and Data Analysis*, 2023.

Zhao H, **Dong Q**. Variable Selection for Additive Hazards Model with Current Status Data. *Journal of Systems Science and Mathematical Sciences*, 2022.

Zhao H, Liu B, **Dong Q**. Jackknife Model Averaging of AFT Model with Current Status Data. *Acta Mathematicae Applicatae Sinica*, 2023.

Chen Z, Lucchesi G, **Dong Q**, Xu Z, Zheng X, Chen Y, Sha M, Cheng W, Ni J, Luo D. Where Does Knowledge Live? Representational, Causal, and Editable Localization in Language Models: A Survey. Preprint under review at TMLR, 2026.

Chen Z, Lucchesi G, **Dong Q**, Zheng X, Song D, Wen Q, Cheng W, Ni J, Luo D. From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens. *Under review*, 2025.

Academic Presentations

Invited Talks

- SILVER Online Lab Meeting Oct 2026
  Department of Epidemiology & Public Health, University of Maryland School of Medicine
- Joint Statistical Meetings (JSM) 2026
- New England Statistics Symposium (NESS) 2026
- ICSA Applied Statistics Symposium 2026

Poster Presentations

- New England Rare Disease Statistics Workshop (NERDS), Boston, MA Oct 2026
- New England Statistics Symposium (NESS) 2026
- The Design and Analysis of Experiments (DAE) 2026
- New England Rare Disease Statistics Workshop (NERDS), Boston, MA Oct 2025
- Dahshu Data Science Symposium, Storrs, CT Oct 2025

Academic Service

Reviewer: *Computational Statistics and Data Analysis*; *Journal of Computational Science*; *Statistical Papers*; *Sankhya B*; *Journal of Systems Science and Mathematical Sciences (Chinese)*

Awards

Student Poster Award, New England Rare Disease Statistics Workshop (NERDS) Oct 2026

UConn Graduate School Conference Participation Award Jul 2026

New England Statistics Symposium (NESS) Poster Award 2026

ICSA Applied Statistics Symposium Student Paper Award 2026

Predoc Fellowship at University of Connecticut 2025

First-class Scholarship at Zhongnan University of Economics and Law 2020, 2022

Technical Skills

Statistics/Methods: survival analysis; interim analysis & conditional power; design simulation; big-data subsampling; model averaging; variable selection

Programming: R (package development), Python (machine learning / deep learning), Git, LaTeX